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Review

Agentic AI and Multi-Agent Collaboration in Healthcare: A Comprehensive Survey of Architectures, Clinical Safety, and Future Directions

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Abstract

In the healthcare and clinical domain, artificial intelligence (AI) is evolving from earlier models that primarily predicted outcomes or generated content toward agentic AI systems that demonstrate the capability to make decisions and complete tasks autonomously. Previous research on AI has contributed significantly to disease identification, deep learning applications, large language models (LLMs), and generative AI. These systems primarily function as assistive tools, as they generate text or predictions without directly interacting with clinical infrastructures. Therefore, recent research trends are increasingly oriented toward agentic AI systems that extend beyond traditional predictive and generative model performance. This manuscript provides a detailed review of the current state of agentic AI, starting with the evolution of AI and the concept of a medical agent. A medical agent refers to an intelligent AI system designed to assist in clinical or administrative tasks by analyzing data, supporting decision making, and interacting with healthcare environments. Its underlying agentic AI architecture integrates planning, memory, reasoning, and environmental interaction to enable autonomous tool use, multi-agent collaboration, and continuous perception decision action loops across diverse healthcare applications and clinical workflows. The review further examines safety mechanisms, including human-in-the-loop oversight, self-verification strategies, and regulatory alignment frameworks, which are designed to ensure reliability, accountability, compliance, and safe deployment in regulated healthcare environments. Our findings indicate that a large number of AI agents have been introduced in various manuscripts for healthcare applications; however, fully autonomous systems remain challenging to achieve, as AI still faces several limitations related to reliability, interpretability, data dependency, and integration within complex clinical workflows. In response to these challenges, this survey shifts the focus from task-specific model performance to system-level autonomy and workflow orchestration, providing a structured foundation for understanding the design, deployment, governance, and limitations of agentic AI systems in modern healthcare ecosystems.

Keywords: agentic AI; multi-agent systems; healthcare safety frameworks; large language model (LLM) agents; human–AI collaboration; clinical workflow automation; AI accountability; health 5.0; system orchestration

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1. Introduction

In today's healthcare ecosystem, clinical facilities increasingly depend on AI to enhance diagnostic accuracy, optimize workflows, and improve patient outcomes. This reliance on

AI is not a recent phenomenon. For several decades, intelligent computational approaches have been continuously used in the medical field to address complex clinical challenges. With rapid advancements in computing power and data availability, modern medical tools are now embedded with AI-driven systems capable of supporting clinicians in decision-making processes [1–3].

1.1. Evolution of AI in Health

The evolution of AI in healthcare can be broadly categorized into three major waves. Figure 1 demonstrates the evolution of AI in health care and clinical uses.

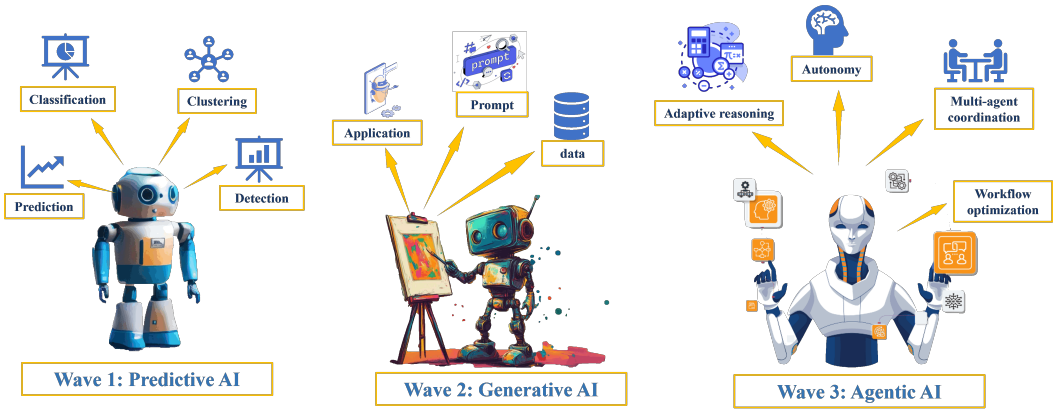


Figure 1. Evolution of Artificial Intelligence in Healthcare: From Predictive and Generative AI to Agentic AI Applications.

The first wave, known as predictive AI, primarily focuses on pattern recognition and outcome prediction using statistical models and deep learning techniques. Predictive AI can accomplish various tasks like classification [4,5], detection [6], identification [7], prediction [8], forecasting [9,10], and clustering [11]. Deep learning model like Convolutional Neural Network (CNNs) have demonstrated remarkable performance in medical imaging tasks such as X-ray [12,13], CT [14], and MRI analysis [15], while Recurrent Neural Networks and Long Short-Term Memory networks have shown exceptional performance in analyzing physiological time-series signals such as ECG [16], EEG [17], heart murmur recordings [18], EMG [19], and PPG signals. These approaches are largely data-driven and rely on datasets to learn correlations between clinical variables and target outcomes. As a result, predictive AI systems are widely adopted for decision support, early diagnosis, and treatment planning in clinical practice.

The second wave of artificial intelligence introduces generative AI, which focuses on content generation, representation learning, and knowledge synthesis. Prominent generative models, including Large Language Models (LLMs) [20,21], GANs [22], and transformer-based generative architectures [23], are capable of learning complex data distributions and generating large volumes of realistic clinical data from trained datasets. These models significantly enhance the performance of deep learning systems by enabling data augmentation, feature enrichment, and improved generalization. Generative AI supports the synthesis of diverse data modalities, such as medical images, physiological signals, and clinical text, thereby contributing substantially to advancements in the healthcare domain. In clinical practice, these systems can process vast amounts of heterogeneous medical data, reducing administrative burden and improving the efficiency of clinical documentation [24]. Despite these advantages, critical challenges including data privacy, model reliability, bias, and the need for rigorous clinical validation remain key considerations for the widespread adoption of generative AI in healthcare applications [25].

The third and emerging wave of artificial intelligence in healthcare is agentic AI, which represents a fundamental shift from predictive and generative paradigms toward autonomous, goal-driven systems capable of acting within complex clinical environments. Agentic AI systems are designed to perceive context, reason over time, plan multi-step actions, and execute decisions across interconnected healthcare workflows. By integrating large language models with mechanisms for planning, memory, tool use, and environmental interaction, it enables direct engagement with hospital information systems, appointment scheduling platforms, billing infrastructures, clinical alerting pipelines, and pre-screening processes [26,27]. Despite the impressive performance of modern generative AI models on medical examinations and clinical reasoning benchmarks, a critical limitation often described as the agency gap remains unresolved [28]. Current LLMs largely function as passive responders; they generate recommendations or textual outputs but lack the capability to autonomously coordinate tasks, adapt to dynamic clinical states, or execute multi-step actions within real-world healthcare systems. Even moderately complex operational tasks that require sequential decision-making, tool interaction, or environmental feedback remain challenging for standalone generative AI models. While prompt engineering and chain-of-thought techniques can partially mitigate these shortcomings, such approaches still rely on explicit human guidance and do not confer true autonomy [29,30]. Figure 2 illustrates key capabilities of predictive, generative, and agentic AI.

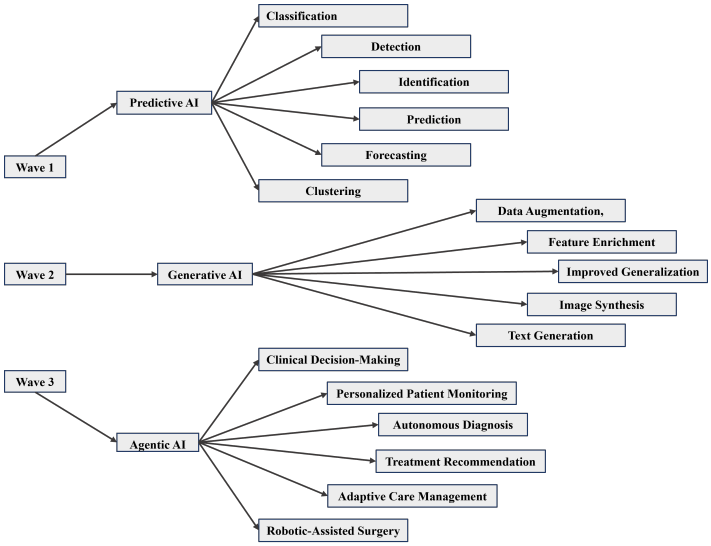


Figure 2. Overview of Core Functional Abilities in Predictive, Generative, and Agentic Artificial Intelligence.

1.2. The "Agency" Gap

Agentic AI has the potential to address this gap by enabling continuous perception–decision–action loops. This enables systems to independently break down high-level clinical goals into actionable steps, monitor outcomes over time, and refine their actions based on feedback [21,31]. Advances in memory architectures and retrieval-augmented reasoning allow agentic systems to maintain long-term contextual awareness and dynamically access up-to-date clinical knowledge. This reduces their reliance on exhaustive task-specific training data [32,33]. Moreover, agent-based reasoning and self-verification mechanisms have been investigated as potential strategies to mitigate common failure modes of generative AI, including hallucinations and inconsistent outputs [26]. These strategies work through step-by-step validation and external knowledge grounding [32,34,35]. Beyond potential improvements in reliability, agentic AI may introduce new efficiencies in computation and deployment, addressing growing concerns surrounding the scalability

and sustainability of increasingly large foundation models. Rather than relying solely on ever-larger architectures trained at substantial computational and environmental cost, agentic systems enable fine-grained control over reasoning depth and resource allocation during inference. Smaller, specialized models equipped with reasoning and planning capabilities can achieve performance comparable to that of larger models by dynamically investing computation where needed, thereby improving cost-efficiency and supporting economically viable deployment even for infrequent or specialized clinical tasks [34,36].

The agency gap in healthcare is really a socio-technical and regulatory barrier rather than just a restriction of thinking or planning abilities, notwithstanding recent technological advancements. Several authors have suggested that the main challenges for the uptake of autonomous medical agents are organizational readiness, fit with clinical workflow, compatibility with legacy electronic health records (EHRs), and lack of standardized protocols for deployment [1,32]. Furthermore, significant uncertainty remains regarding accountability and legal responsibility when autonomous systems produce harmful recommendations or actions. The lack of clear liability frameworks raises unresolved questions regarding whether responsibility should rest with the developer, clinician, healthcare institution, or AI vendor. Existing regulatory frameworks also favor human-in-the-loop oversight for high-risk clinical activities, including medication management and procedure scheduling, thereby limiting the degree of autonomy that can be safely delegated to AI agents [25]. Consequently, achieving genuine agentic autonomy in healthcare will require the development of governance-aware architectures, transparent decision-logging mechanisms, clearly defined accountability frameworks, and sustained cross-disciplinary collaboration among clinicians, regulators, policymakers, and AI developers [32,33,37,38].

1.3. Objectives and Contributions

Recent survey studies have made significant progress in exploring agentic AI for healthcare. However, most existing studies fail to provide a complete and broad summary of the existing concepts. Additionally, these papers contain certain limitations and research gaps that motivate the present work. Table 1 illustrates a comprehensive survey of recent review papers in the field of agentic AI in healthcare. This review paper aims to provide a structured and comprehensive overview of agentic AI in healthcare, clearly defining its scope, capabilities, and boundaries to facilitate understanding among researchers, clinicians, system developers, and policy-makers. By moving beyond conventional predictive and generative AI paradigms, this survey highlights how agentic systems enable autonomous reasoning, workflow integration, and human–AI collaboration within clinical environments. The key contributions of this review are summarized as follows:

Table 1. Comparison of Existing Surveys and Research Gaps in Agentic AI and Medical AI Systems.

Study	Focus Area	Medical Agents	Multi-Agent Systems	Taxonomy	Action Analysis	Limitations and Research Gaps
[39]	Survey on agent-based healthcare frameworks	N	Y	N	Y	Lacks modern LLM-based agents, a unified taxonomy.
[40]	LLM-based agentic AI survey	Y	Y	Y	Y	No deployment-scale engineering (real-world integration, latency, scalability).
[26]	Review of agent vs. base-LLM performance	Y	Y	N	Y	Outcome heterogeneity prevented meta-analysis.
[41]	Digital medicine & teammates	Y	Y	N	Y	Lacks systematic technical benchmarking and architectural depth.
[35]	AI Agents vs. Agentic AI	Y	Y	Y	Y	Lacks specialization in clinical safety.

Table 1. *Cont.*

Study	Focus Area	Medical Agents	Multi-Agent Systems	Taxonomy	Action Analysis	Professional Limitations and Research Gaps
[26]	AI Agents in Clinical Medicine	Y	Y	N	Y	Limited discussion on non-clinical operations.
[21]	LLM Agents in Medicine	Y	Y	N	N	Limited to English-language text-based LLMs and lacks multimodal agentic loops.
[36]	Survey of LLMs	Y	N	N	N	Fails to address domain-specific medical execution.
[33]	Agentic AI Overview	Y	Y	N	Y	Less focus on post-GenAI clinical integration and deployment challenges.
[42]	Agentic AI for Science	N	Y	Y	Y	Overlooks real-time clinical care and bedside integration.
[43]	Autonomous LLM Agents	N	Y	N	Y	Lacks medical safety-critical constraints.
[44]	Potential of LLM Agents	N	Y	N	Y	Lacks medical decision–action integration.
[45]	Agentic LLMs Survey	N	Y	N	N	Lacks a framework for medical tool interaction.
[46]	Agentic RAG	Y	Y	N	Y	Limited to retrieval-augmented patterns and lacks analysis of clinical goal drift.
[47]	Governance of Agentic AI	N	Y	N	N	No discussion of clinical workflow integrity risks.
[21]	Agentic Programming	N	Y	Y	Y	Ignores healthcare-specific ethics and risks.
[48]	Time Series Reasoning	Y	Y	Y	N	Lacks general clinical reasoning and dialogue analysis.
[25]	EMS EHR Automation	Y	N	N	Y	Narrow scope limited to emergency documentation.
[49]	LLM to LMM Literature	N	N	Y	N	Ignores multi-agent collaboration and orchestration.
[23]	BioGPT Research	Y	N	N	N	Lacks comparative survey analysis of healthcare agents.
[3]	Consumer Perspectives	Y	N	N	N	Survey of patient perception; predates modern agentic AI architectures.
[20]	Systematic Review of Evaluation	Y	N	N	N	Lacks discussion of autonomous action loops.
[32]	Agentic AI in Healthcare	Y	Y	N	N	Limited to representative healthcare AI models.

- Clinical Agency Definition: A formal conceptualization of medical agents that moves beyond the "AI as a chatbot" paradigm, defining systems capable of autonomous tool use, multi-step reasoning, and interaction with clinical infrastructures such as electronic health records and diagnostic services.
- Workflow-Specific Taxonomy: A healthcare-centered taxonomy that categorizes agentic AI systems according to clinical and administrative workflows, such as triage, diagnosis, and operational management, thereby directly linking agentic capabilities to real-world hospital challenges.
- Doctor-In-The-Loop Responsibility: Analysis of methods for keeping the control and responsibility of clinicians in the process of agentic workflows, looking at when and how the agentic system needs to take the doctor's opinion into account.
- Clinical and Operational Applications: A synthesis of current and emerging use cases of agentic AI across healthcare settings, illustrating how agentic systems extend beyond static decision support toward intelligent orchestration of clinical and administrative processes.

- Challenges and Open Issues: An analysis of key technical and practical challenges, including goal specification, adaptive context handling, reliability, and integration with resource-constrained healthcare environments.
- Future Research Directions: Actionable recommendations for future research, focusing on scalable architectures, safe deployment, ethical governance, and the integration of agentic intelligence into regulated clinical workflows.

Overall, this paper goes beyond a conventional literature survey by offering a well-organized and application-driven foundation for understanding, developing, and responsibly deploying agentic AI systems in healthcare, with an emphasis on clinical safety, operational effectiveness, and ethical sustainability.

1.4. Distribution of Collected Literature Across Databases and Study Types

A systematic literature search was conducted following PRISMA 2020 guidelines to ensure transparency and reproducibility. The search was performed between 24 December 2025 and 3 February 2026 across multiple databases, including IEEE Xplore, PubMed, Google Scholar, ACM Digital Library, Springer, Elsevier, arXiv/preprint servers, and ResearchGate.

The following Boolean search strings were used with appropriate adaptations across databases:

- ("AI" OR "Artificial Intelligence" OR "Agentic AI") AND ("AI Agent" OR "Single Agent" OR "Multi-Agent Systems")
- AND ("Healthcare" OR "Medical" OR "Clinical Use" OR "Medicine")
- AND ("LLM" OR "Large Language Model" OR "Medical Assistant" OR "Radiology Agent" OR "Clinical Assistant")
- AND ("Application" OR "Architecture" OR "Autonomy" OR "Decision Support" OR "MRI" OR "ECG" OR "EEG" OR "X-ray")

Only English-language peer-reviewed journal articles, conference papers, and preprints with sufficient methodological detail were included. Editorials, opinion pieces, non-healthcare applications, and non-technical reports were excluded.

The initial search yielded approximately 743 records. After removal of duplicates (n = 52), 691 records remained for screening. Title and abstract screening excluded 384 records due to irrelevance to healthcare-focused agentic AI applications. A total of 307 full-text articles were assessed for eligibility.

During full-text screening, 207 articles were excluded for the following reasons:

- Non-healthcare domain (n = 26)
- Absence of agentic AI relevance in healthcare context (n = 94)
- Insufficient technical or methodological details (n = 49)
- Conference abstracts without full paper availability (n = 38)

Finally, 100 studies were included in the qualitative synthesis.

All screening and selection processes were initially conducted by the first author. To reduce potential selection bias, inclusion and exclusion decisions were guided by predefined eligibility criteria, and ambiguous cases, together with a subset of the screening decisions, were subsequently reviewed by the second author and the project instructor. Any recommendations arising from this secondary review were incorporated, and the final study selection and manuscript revisions were completed by the first author.

Figure 3 presents the PRISMA flow diagram summarizing the study selection process and literature screening stages.

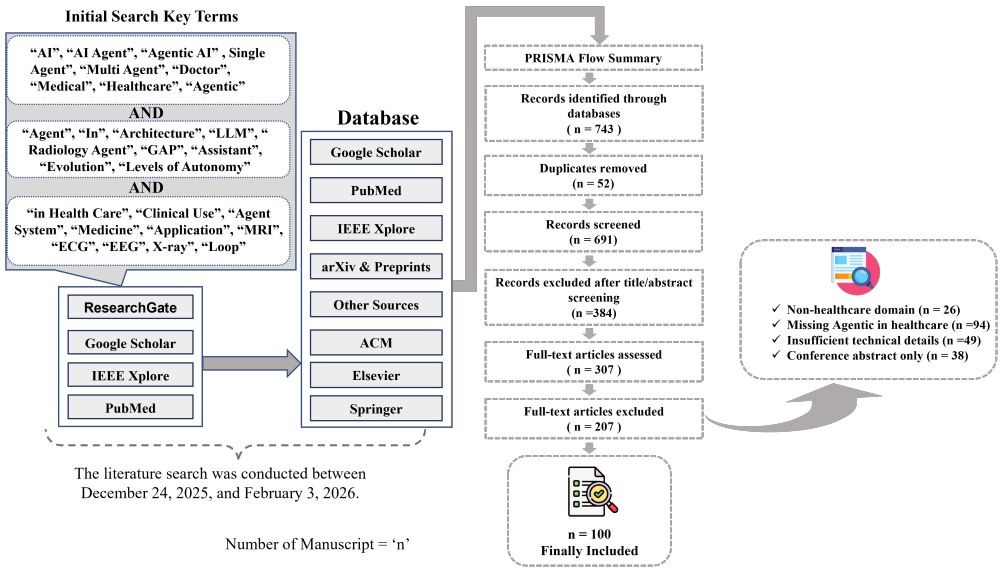


Figure 3. PRISMA flow diagram detailing the systematic literature search and study selection process.

The dataset consists of 100 research papers, primarily indexed in IEEE, Springer, Elsevier, PubMed, arXiv, MDPI, and ACM, with 39 grouped as mixed/other sources due to overlapping or incomplete metadata. The literature shows strong reliance on Scopus-indexed venues, with significant contributions from IEEE, Springer, and Elsevier. Table 2 presents the distribution of the collected papers across major databases and sources, including IEEE Xplore, Springer, Elsevier, PubMed, arXiv, MDPI, ACM, and other sources, totaling 100 papers.

Table 2. Number of papers by major databases/sources (Total = 100).

Database/Source	References	Number of Papers
IEEE Xplore	[5,17–19,33]	5
Springer	[2,15,25,43,50–56]	11
Elsevier/ScienceDirect	[1,4,6,7,12,37,41,57–60]	11
PubMed/Biomedical	[3,20,27,61–63]	6
arXiv/Preprints	[21,26,29–31,34–36,42,48,64–73]	20
MDPI/Open Access	[22,38,74,75]	4
ACM Digital Library	[76–79]	4
Other/Mixed/Multi-indexed Sources	(Additional interdisciplinary journals, overlapping indexing, hybrid venues, and uncategorized references across databases)	39
Total		100

Figure 4 shows the distribution of the collected papers by publication year. A considerable portion of studies from arXiv reflects the rapidly evolving nature of agentic AI research. PubMed-indexed works highlight strong healthcare relevance, while MDPI and ACM add interdisciplinary and open-access perspectives. Figure 5 illustrates the same dataset visually, showing the proportional database-wise distribution of the collected references.

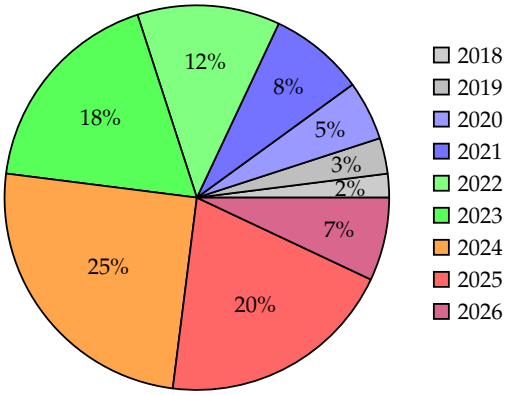


Figure 4. Publication year distribution of the collected 100 papers (2018-2026).

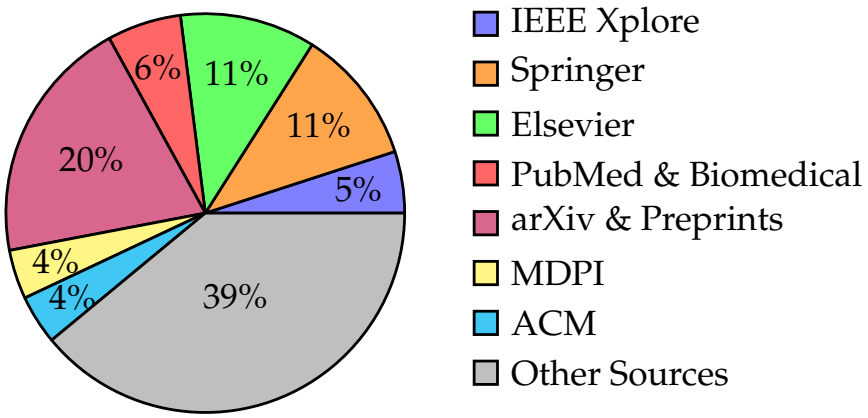


Figure 5. Database distribution of collected references (Total = 100 papers).

From a study-type perspective, the collection includes surveys, technical studies, and agentic AI-focused research, with surveys providing synthesis and technical works contributing methodological advances. A growing subset focuses on LLM-based agentic systems, indicating rapid expansion of this research area.

An analysis of the excluded literature (Table 3 (Part-A) and Table 4 (Part-B), Clinical Risk, Evaluation Method, Deployment Status, and Study-Type Classification) reveals a stark structural imbalance in contemporary healthcare AI research. Approximately 65% of existing publications are limited to conceptual taxonomies or literature surveys, while 80% remain entirely theoretical or research-only without any functional prototype instantiation. This distinct lack of active, non-survey architectural frameworks highlights the critical need for technical evaluations of implemented agentic systems. Table 5 summarizes the key characteristics of agentic AI systems, including autonomy level, agent architecture, tool use, memory, retrieval-augmented generation (RAG), human-in-the-loop integration, and safety mechanisms across the reviewed studies.

Table 3. (Part A) Clinical Risk, Evaluation Method, Deployment Status, and Study-Type Classification. Abbreviations: RO = Research-only; P = Prototype; C = Conceptual; R = Review; SU = Survey; BS = Benchmark Study; SI = Simulation.

Study	Clinical Risk	Evaluation Method	Deployment Status	Study Type
Mennella et al. (2024) [1]	High	Narrative Review	RO	C/SU
Gridach et al. (2025) [42]	High	Literature Survey	RO	C/SU
Esmaeilzadeh (2020) [3]	Low	Consumer Survey Study	RO	C/SU
Heidari et al. (2020) [12]	Medium	Benchmark (Chest X-Ray)	RO	BS

Table 3. *Cont.*

Study	Clinical Risk	Evaluation Method	Deployment Status	Study Type
Al-Yasriy et al. (2020) [14]	Medium	Benchmark (CT Scan Datasets)	RO	BS
Hardaha et al. (2023) [15]	Medium	CNN MRI Literature Survey	RO	C/SU
Shokrollahi et al. (2023) [69]	High	GenAI Literature Survey	RO	C/SU
Aljohani et al. (2022) [22]	Low	Synthetic Images Benchmark	RO	SI
Shool et al. (2025) [20]	High	LLM Evaluation Systematic Review	RO	C/SU
Al-Kateb et al. (2025) [23]	High	Genomic Dataset Benchmark	RO	P
Bai et al. (2025) [25]	Medium	Simulation-based User Study	P	S
Sallam (2023) [29]	High	ChatGPT Systematic Review	RO	C/SU
Nori et al. (2023) [30]	High	USMLE Benchmark Study	RO	BS
Matarazzo et al. (2025) [36]	Medium	LLM Literature Survey	RO	C/SU
Schneider (2025) [34]	High	Gen-to-Agentic Conceptual Survey	RO	C/SU
Karunanayake (2025) [37]	High	Literature Review	RO	C/SU
Li et al. (2024) [50]	Medium	Literature Review	RO	C/SU
Chang et al. (2025) [48]	Medium	Literature Survey	RO	C/R
Huang et al. (2024) [49]	Low	Literature Review	RO	C/R
Sapkota et al. (2025) [35]	Medium	Taxonomy Analysis	RO	C/R
Chowdhury et al. (2025) [70]	High	Framework Evaluation	P	BS
Begum et al. (2025) [51]	High	Framework Conceptualization	RO	C/SU
Thapa et al. (2025) [57]	High	Zero-shot Accuracy Audit	RO	BS
Wang et al. (2024) [43]	Medium	Multi-modal Benchmark	P	BS
Dharmale et al. (2023) [80]	Low	Sensor Validation	P	P
Zaretsky et al. (2024) [61]	Medium	Physician Evaluation Study	P	SI
Janota & Janota (2025) [52]	High	Qualitative Cohort Study	P	P
Koçak & Meşe (2025) [81]	Medium	Literature Survey	RO	C/SU
Suura (2025) [82]	High	Framework Conceptualization	RO	C/R
Salehi et al. (2025) [75]	Medium	Ethical Review	RO	C/SU
Miehling et al. (2025) [31]	Medium	Systems Theory Metaview	RO	C/R
Welch et al. (2025) [71]	Low	Dialogue Pipeline Evaluation	P	P
Hong et al. (2022) [62]	Medium	Feasibility Survey	P	P
Schmidgall et al. (2024) [72]	High	Multimodal Simulation Testing	RO	BS

Table 4. (Part B) Clinical Risk, Evaluation Method, Deployment Status, and Study-Type Classification. Abbreviations: RO = Research-only; P = Prototype; C = Conceptual; R = Review; SU = Survey; BS = Benchmark Study; SI = Simulation; A = Analysis.

Study	Clinical Risk	Evaluation Method	Deployment Status	Study Type
Almutairy et al. (2025) [83]	Low	Statistical Impact Analysis	RO	BS
Gu et al. (2024) [76]	Low	Wizard-of-Oz Study	RO	P
Phillip & Joshua (2025) [84]	Low	Conceptual Synthesis	RO	C/R
Grange et al. (2025) [77]	High	Process Orchestration Modeling	RO	SI
Huang (2025) [53]	High	Textbook Overview	RO	C/R
Ko et al. (2025) [85]	Low	Educational Pilot Trial	P	P
Li et al. (2025) [78]	Medium	Cross-sectional Study	RO	BS
Fan et al. (2025) [86]	High	Multi-agent Benchmark	RO	BS
Bullock et al. (2019) [13]	Medium	Pixel Accuracy Validation	RO	BS
Feng et al. (2025) [73]	Low	Software Taxonomy Analysis	RO	C/SU
Festor et al. (2021) [54]	High	Clinical Risk Tiering	RO	A
Vallor & Vierkant (2024) [28]	Low	Epistemic Philosophy Review	RO	C/R
Yue et al. (2020) [24]	Medium	Algorithmic Survey	RO	C/S
Wang et al. (2025) [21]	High	Medical Agent Survey	RO	C/S
Zou & Topol (2025) [63]	High	Clinical Commentary	RO	C/S
Milne-Ives et al. (2020) [27]	Medium	Systematic Review	RO	C/R
Gorenshtein et al. (2025) [26]	High	Systematic Review	RO	C/R
Iliashenko et al. (2019) [87]	Low	Exploratory Literature Scan	RO	C/R
Aung et al. (2021) [88]	Low	Healthcare Survey	RO	C/SU
Khalil et al. (2025) [89]	Medium	Eldercare Taxonomy Survey	RO	C/SU
Brohi et al. (2025) [38]	Medium	Cross-domain Mapping	RO	C/SU
Dwivedi et al. (2026) [90]	Low	Tourism Survey	RO	C/R
Procter et al. (2023) [79]	High	Ethnographic Critique	RO	C/R
Wang et al. (2025) [91]	Low	Software Engineering Survey	RO	C/SU
Banerjie et al. (2025) [32]	High	Taxonomy Mapping	RO	C/SU
Acharya et al. (2025) [33]	Low	Tech Survey	RO	C/SU
Jain & Singh (2018) [4]	Medium	Feature Extraction Review	RO	C/R
Alhassan & Zainon (2021) [5]	Medium	Dimensionality Reduction	RO	C/SU
ALEnezi (2019) [6]	Medium	GLCM Benchmark Test	P	BS

Table 4. *Cont.*

Study	Clinical Risk	Evaluation Method	Deployment Status	Study Type
Vayadande et al. (2024) [7]	Medium	Classifier Review	RO	R
Alkhanbouli et al. (2025) [8]	Medium	Explainable AI Review	RO	C/R
Merabet et al. (2024) [9]	High	IoMT System Review	RO	C
Jamshidi et al. (2022) [10]	Low	Forecast Review	RO	R
Loftus et al. (2022) [11]	Medium	Phenotype Clustering Review	RO	R
Biswas et al. (2025) [16]	High	MIT-BIH Validation	RO	BS
Kamson et al. (2023) [18]	Medium	PhysioNet Challenge	RO	BS
Tripathy et al. (2025) [17]	High	Seizure Dataset Audit	RO	BS
Inam et al. (2021) [19]	Low	EMG Review	RO	C
Shavit et al. (2023) [47]	Low	Governance Analysis	RO	C/R
Singh et al. (2025) [46]	Low	Agentic RAG Survey	RO	C/SU
Plaat et al. (2025) [45]	Low	LLM Agent Survey	RO	C/SU
Xi et al. (2025) [44]	Low	Capabilities Survey	RO	C/R
Xu et al. (2026) [39]	High	Healthcare AI Agent Review	RO	C/R
Vatsal et al. (2026) [40]	High	Empirical Taxonomy Study	RO	C /R
Gorenshtein et al. (2025) [26]	High	Clinical Agent Review	RO	C/R

Table 5. Categorization of Healthcare AI Research Along the Translation Spectrum: System Maturity, Validation Environments, and Patient Safety Risk Profiles.

Study	Autonomy Level	Agent Architecture	Tool Use	Memory	RAG	Human-in-the-Loop	Safety Mechanism
Li et al. (2024) [64]	Semi-autonomous	Single	external domain tools	Short-term	In-context learning datasets	user prompt entry	Instruction tuning-based validation rules
Zeggai et al. (2025) [65]	Semi-autonomous	Single	Vaccine schedule engines	Short-term	Vectorized CDC Guidelines	Patient entries	Grounding guardrails on medical advice
Kim et al. (2024) [92]	Semi-autonomous	Multi	Clinical calculators, diagnosis tools	Short-term	In-context clinical guidelines	Physician verification	Consensus validation & ensemble voting
Li et al. (2024) [66]	Autonomous	Multi	Digital EHR databases, tests	Long-term	Internal medical knowledge base	Simulated environment	Automated evaluator rewards & penalty rules
Humayun et al. (2022) [74]	Autonomous	Multi	IoT Health sensors, APIs	Short-term	No	Automated escalation alerts	Hardcoded threshold rule sets
Borque-Benedí et al. (2025) [93]	Semi-autonomous	Multi	No	Short-term	No	User prompts	Prompt-based instruction guardrails
Tian et al. (2025) [67]	Autonomous	Multi	prompt refinement	Short-term	No	Automated optimization	Multi-agent critique and consensus checks

Table 5. *Cont.*

Study	Autonomy Level	Agent Architecture	Tool Use	Memory	RAG	Human-in-the-Loop	Safety Mechanism
Liu et al. (2025) [68]	Semi-autonomous	Single	SDMSD Defect Detector, EGR Tool	Short-term	NDT domain corpus	User visual queries	Evidence-Grounded Reflection (EGR) tool check
Yan et al. (2026) [94]	Autonomous	Multi	laboratory and imaging tools simulated	Long-term	No direct RAG integration	human expert evaluations for assessment	Data privacy protection, ethical review, and controlled dataset distribution
Ruiz Mejia & Rawat (2025) [95]	Autonomous	Multi	EHR schedule matrices, databases	Long-term	Provider database embeddings	Final scheduling alerts	Profile matching constraint enforcement
Vennaro (2024) [96]	Semi-autonomous	Single	Cloud Office APIs, Microsoft Graph	Short-term	Administrative vector databases	Staff approvals	Role-Based Access Control cloud keys

2. Defining the Medical Agent

The concept of a medical agent is a system integrated with artificial intelligence that operates within clinical environments to support healthcare delivery, decision-making, documentation, and workflow coordination [64]. Unlike conventional AI models that only generate responses, a medical agent is capable of perceiving patient data, reasoning over clinical context, invoking external medical tools, and taking task-oriented actions within healthcare systems [37].

2.1. Core Components

Medical agents can be understood through the same foundational components used in agentic AI systems: perception, brain, tools, memory, and action. These components collectively enable perception–reasoning–action loops similar to human clinical cognition. Figure 6 shows the core components of the medical agent architecture.

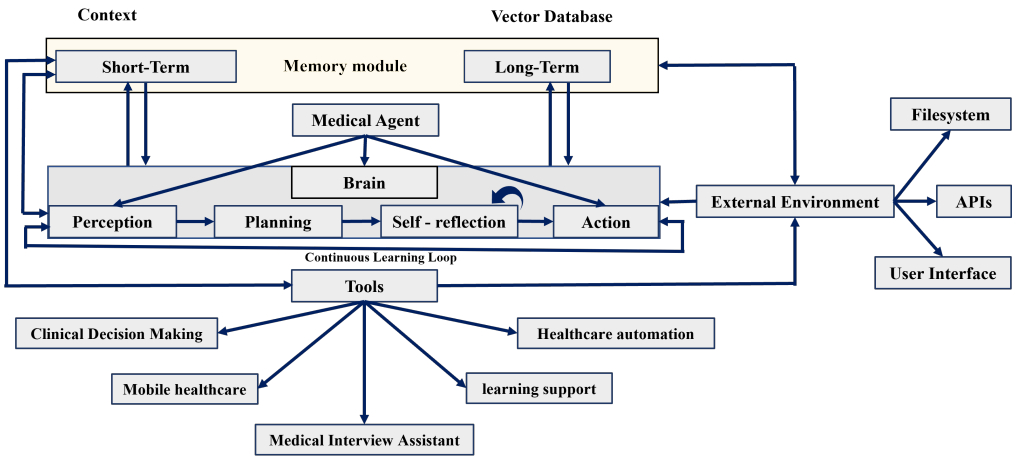


Figure 6. Core Component of Medical Agent Architecture.

- Perception: Multimodal Data Acquisition
  Perception means that the medical agent can simultaneously process multiple types of clinical information rather than relying on a single data source. It can process unstructured data, like clinical notes that are written by physicians, nurses, or specialists and that contain patient history, chief complaints, differential diagnoses, and impressions. It can process laboratory results and structured numeric data. The agent can recognize patterns, trends, and threshold violations [58]. It can process visual data, X-ray,

MRI, CT, and ultrasound images. The agent may flag suspected tumors, fractures, hemorrhage, or subtle anomalies that warrant clinician review [50]. It can process time-series data like heart rate, blood pressure, respiratory rate, temperature, and oxygen saturation. Continuous or high-frequency measurements allow early detection of deterioration [48]. The medical agent advances from basic text processing to patient-centric situational awareness through multimodal perception. This concept is also reflected in multimodal IoT-enabled emergency and response care systems, where physiological signals from biosensors and medical IoT devices are integrated with imaging data such as X-rays, MRI, and CT scans to support real-time clinical decision-making [97]. However, because sensitive patient data is handled across various modalities, it also presents possible security and privacy threats. These concerns are further addressed in the Privacy, Security, and Attack Surface Expansion subsection.

- Brain: Clinical Reasoning and Planning
  The "brain" of the medical agent relies primarily on large foundation models, including large language models (LLMs) and large multimodal models (LMMs) [31,67,68]. In agentic architectures, the model acts as a central coordination hub. It interprets clinician commands, decomposes complex tasks, calls external tools, evaluates outputs, and maintains logical consistency. These models perform medical reasoning in chain-of-thought style, conduct clinical thinking, generate differential diagnoses, formulate plans, interpret guidelines, develop summaries, and assist with documentation [35,49].
- Tools: The Differentiator
  Tools provide a bridge between reasoning and clinical action. The defining feature that distinguishes an agent from a chatbot is tool use. Medical agents can invoke a wide range of external tools, including biomedical literature databases (PubMed, clinical guidelines) [65,70], update clinical knowledge portals, hospital electronic health records [51], system FHIR APIs for structured health data access, medical calculators (e.g., eGFR, CHA_2DS_2-VASc, ASCVD risk) and drug–drug interaction checkers [57], and scheduling and messaging systems. Through tools, the agent performs operational steps, not merely conversation.
- Operational Output
  In clinical workflows based on perception and reasoning, and supported by tools, the medical agent produces actionable outputs such as drafting clinical notes or discharge summaries, sending alerts for abnormal vitals or laboratory trends, preparing order sets or suggestions for clinician approval, booking appointments or follow-up slots, and triaging messages or prioritizing patient queues. Importantly, actions are assistive, not an autonomous replacement of clinicians. The agent prepares, drafts, or recommends while the clinician validates, signs, and remains accountable.

2.2. Levels of Autonomy in Healthcare Agents

Autonomy in healthcare AI agents describes the degree to which an intelligent system can perform clinical tasks independently of human involvement while still ensuring safety, accountability, and ethical integrity. Drawing on the conceptual foundations of Festor et al. [54] and Feng et al. [73], healthcare AI autonomy is organized into five progressive levels. These levels differ mainly in terms of (i) the clinician's role, (ii) how decisions are produced and executed, and (iii) where responsibility ultimately lies. Figure 7 shows levels of autonomy in healthcare agents and the role of human involvement.

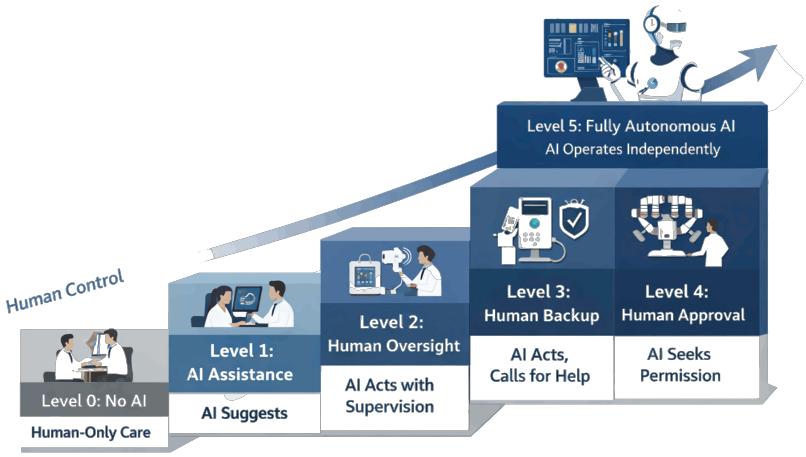


Figure 7. Human-AI Interaction Across Levels of Autonomy in Healthcare Systems.

2.2.1. Level 0: No AI (Human-Only Care)

Level 0 represents the baseline where no artificial intelligence is used. All diagnostic, therapeutic, and operational decisions are made exclusively by healthcare professionals. This level provides a reference point against which the performance, safety, and reliability of AI-enabled systems can be evaluated.

Example: Conventional clinical practice without decision support tools.

2.2.2. Level 1: AI as a Suggestive Assistant

At Level 1, the AI agent offers recommendations or insights but does not act in the clinical environment. The clinician retains full control and responsibility, deciding whether or not to follow the AI's advice. Here, the AI functions as a supportive "copilot" that enhances awareness and efficiency without replacing human judgment.

Example: An AI system that flags abnormal lab results and suggests possible diagnoses while the physician makes all final decisions.

2.2.3. Level 2: AI with Continuous Human Oversight

At Level 2, the AI is allowed to execute actions, but under constant human supervision. The clinician continuously monitors the agent's outputs and can intervene or override decisions at any moment. This level emphasizes close collaboration between human and AI.

Example: An insulin infusion system that automatically adjusts dosage while clinicians continuously supervise patient status.

2.2.4. Level 3: AI with Conditional Human Backup

Level 3 systems operate autonomously for extended periods without continuous monitoring. However, they are designed to recognize uncertainty and request human input when confidence is low or when unexpected situations arise. The clinician acts as a consultant, providing feedback and guidance rather than direct control.

Example: A ventilator that adapts to patient breathing patterns and alerts clinicians when abnormal conditions are detected.

2.2.5. Level 4: AI with Human Approval Only

At Level 4, the AI agent performs most tasks independently and only involves clinicians when approval is required for high-impact or sensitive actions. The human role becomes largely supervisory and confirmatory.

Example: An AI system that manages medication schedules and seeks approval only for high-risk drug changes.

2.2.6. Level 5: Fully Autonomous AI

Level 5 represents complete autonomy. The AI agent plans, executes, and adapts all actions without any user involvement. Clinicians can only observe system behavior through logs or dashboards and may have access to an emergency stop mechanism. Due to ethical, legal, and safety concerns, this level remains largely theoretical in healthcare.

Example: A fully autonomous clinical agent that diagnoses and treats patients without clinician input.

2.2.7. Discussion: Balancing Autonomy and Risk

In healthcare, where decisions have direct consequences for patient well-being, autonomy must be carefully aligned with task complexity, risk level, and accountability, because higher autonomy does not automatically imply better or safer systems. Most current real-world healthcare AI systems operate between Level 1 and Level 3. Fully autonomous agents (Level 5) raise significant concerns regarding trust, transparency, and liability.

Therefore, effective healthcare agents should strike a balance between automation and human oversight, using AI to enhance efficiency and consistency while ensuring that clinicians remain meaningfully involved in high-stakes decision-making. From a socio-technical standpoint, professional autonomy remains a foundational element of safe and ethical medical practice. Polder and Jochemsen argue that healthcare systems must preserve essential physician autonomy to sustain trust in the patient–provider relationship while balancing efficiency and governance [55]. Empirical evidence further shows that increased system complexity often necessitates cooperation mechanisms that can unintentionally reduce perceived professional autonomy, particularly in tightly coupled multidisciplinary workflows [56]. These findings suggest that healthcare agents should be designed to support consultation and coordination without displacing clinical authority.

Consequently, medical agents represent a practical and ethically grounded pathway toward doctor-supervised autonomy rather than doctor replacement. By perceiving multimodal clinical data, reasoning over uncertain information, invoking domain-specific tools, and executing workflow-level actions, such agents can substantially reduce administrative burden and improve operational efficiency. However, consistent with the autonomy frameworks discussed above, physicians must remain the final decision-makers, with agents functioning as drafting, monitoring, and orchestration systems that enhance rather than erode human clinical judgment.

3. Architectures & Frameworks

The architecture of autonomous systems can broadly be categorized into AI agents and agentic AI. AI agents typically involve a single autonomous agent designed to perform modular, tool-assisted tasks within a defined scope. In contrast, agentic AI refers to multi-agent systems where multiple interacting agents coordinate, plan, and collaborate to accomplish complex, distributed objectives [35]. Figure 8 illustrates this distinction, where the medical AI agent represents the single-agent paradigm, and the multi-agent configuration corresponds to agentic AI.

3.1. Single-Agent Systems

Single-agent systems use a single large language model that reasons over patient data and produces outputs such as intake conversations, clinical summaries, reports, and discharge documentation. Using ReAct prompting, the agent alternates between reasoning steps and tool-use actions under medical guidelines, enabling structured history-taking and accurate summarization. Such systems align well with clinical workflows where the

goal is to translate complex medical information into accurate, readable, patient-friendly documentation while maintaining safety through clinician oversight.

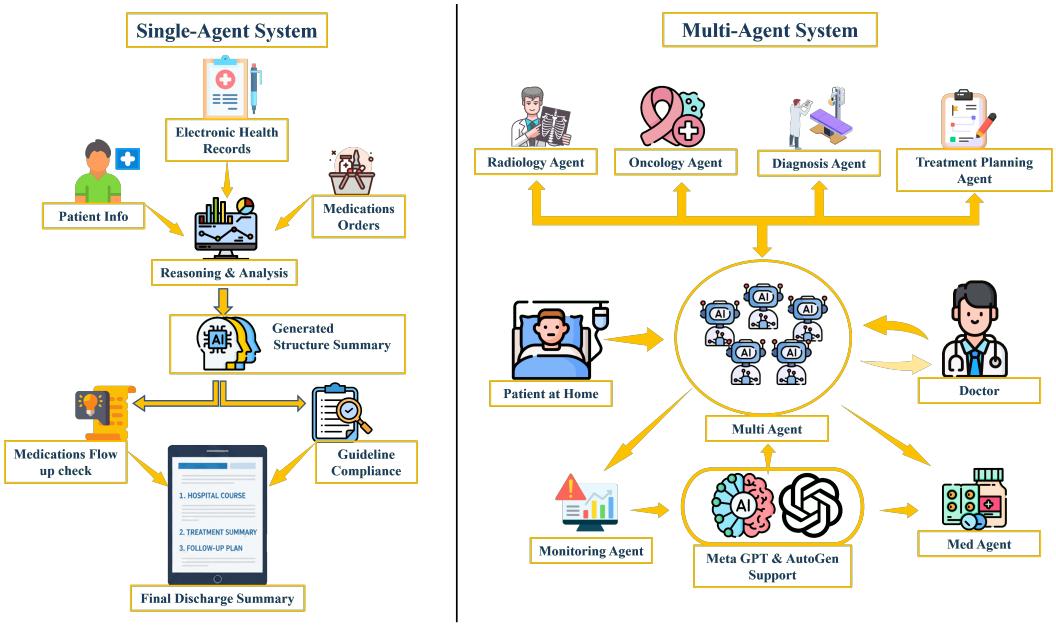


Figure 8. System Architectures of Single-Agent and Multi-Agent AI-Based Healthcare Frameworks.

Wang et al. [43] show that LLMs enhance computer-aided diagnosis (CAD) outputs such as diagnosis, lesion segmentation, and report generation by summarizing information in natural language. The study demonstrated improved diagnostic performance and the creation of patient-friendly interactive systems compared to conventional CAD systems. Kim et al. [92] demonstrate how LLMs can support medical decision-making tasks of varying complexity and show strong performance on multiple clinical benchmarks. They highlight that combining reasoning mechanisms and external medical knowledge leads to significant accuracy improvements, supporting structured decision processes. Li et al. [66] introduce a simulacrum environment that enables agents to treat large numbers of virtual patients and evolve decision-making abilities without manual labeling. Doctor agents trained in this simulated hospital outperformed state-of-the-art medical agent methods on USMLE-style test questions. Dharmale et al. [80] develop an intelligent system that provides automated medication reminders and supports interaction between patients and doctors. It emphasizes cloud storage of medical information, image processing for prescriptions, and secure data handling, encouraging adherence to treatment. AI-generated summaries were found to be more efficient and coherent than clinician-written summaries. Studies report improvements in information structure and preparation time. In this field, LLMs significantly reduced reading grade level and improved the understandability scores of discharge summaries [52,61].

3.2. Multi-Agent Systems

Multi-agent systems (MASs) in healthcare function like a virtual hospital team, mirroring real multidisciplinary consultations such as tumor boards. MAS applications can allow patients to remain at home while reducing healthcare costs and are important for aging populations and increasing service demand [41]. In this framework, multiple AI agents adopt specialized clinical roles and collaborate across functions including image interpretation, diagnosis, treatment planning, and follow-up care, thereby moving beyond single-model AI toward distributed reasoning and expertise [63,82].

For example, a radiology agent may interpret imaging scans and report findings, which are then evaluated by an oncology agent against clinical guidelines to propose management options before being presented to a human physician for final decision-making and oversight [81]. This type of workflow reflects a broader shift in radiology from single AI systems toward multi-agent networks that are capable of autonomous reasoning and coordinated decision-making [75]. As these agents interact and build upon one another's outputs, the authors introduce the concept of compound opacity, wherein such interactions generate new challenges for transparency and accountability in radiological workflows. Li et al. [64] introduces a multi-modal medical agent capable of selecting appropriate specialized medical tools for given tasks named MedAgent. The agent is shown to perform medical tasks across multiple modalities and is efficient in updating and integrating new tools, demonstrating adaptability in clinical contexts. The study by Humayun et al. [74] presents a multi-agent health monitoring system that collects patient data, reasons over it, and recommends actions to patients and clinicians. It is particularly suited for chronic, aged, and remote patients, offering efficient monitoring beyond confined hospital environments.

Some other technologies, such as MetaGPT and AutoGen [93], support this approach by enabling role-based agent assignment, structured inter-agent dialogue, and medical tool or knowledge retrieval within real clinical decision-making scenarios.

3.3. Analytical Synthesis of Agent Architectures in Healthcare: Single-Agent vs. Multi-Agent Systems

The architectural contrariness between Single-Agent Systems and Multi-Agent Healthcare Systems reflects a fundamental transition from task-specific automation to orchestrated clinical agency. Single-agent architectures prioritize functional modularity, employing LLMs to orchestrate tools for discrete, deterministic tasks such as clinical calculations or note summarization. While efficient, these modular entities are limited by "task brittleness" and episodic memory. Conversely, multi-agent systems utilize "collaborative reasoning" ecosystems, where specialized agents simulate multidisciplinary care teams to tackle complex, cross-disciplinary challenges like rare disease diagnosis. Although MAHS architectures offer superior multi-modal integration and persistent context sharing, they introduce "coordination overhead" and risks of "error cascades." Empirical evidence suggests performance optimization occurs at a threshold of 4–5 agents, beyond which coordination complexity yields diminishing returns. Future clinical integration necessitates a tiered framework where system autonomy is precisely calibrated to task complexity to maintain safety and accountability [21,26,32,35]. Table 6 shows a comparative overview of single-agent vs. multi-agent systems in healthcare.

The deployment of agentic AI in urgent medical situations introduces significant latency and operational challenges, as the transition from passive prediction to continuous sense–think–act cycles requires high-speed multimodal data integration [32,37]. In emergency triage, multi-agent systems must rapidly process real-time streams of vital signs, symptoms, and medical history to prioritize high-risk cases; however, resource contention and coordination overhead can create systemic bottlenecks that delay clinical responsiveness [25,35]. Similarly, autonomous ICU monitoring for conditions like sepsis risk relies on persistent memory and reasoning engines that must balance diagnostic precision with the computational demands of long-horizon data processing [21,35]. To address these constraints, edge computing and Small Language Models (SLMs) have emerged as critical strategies, enabling real-time AI inference directly on medical devices to reduce reliance on centralized cloud infrastructure while maintaining the speed necessary for life-saving interventions in resource-limited environments [34,37]. Furthermore, repeated LLM inference cycles and inter-agent communication may increase end-to-end response times, creating a trade-off between diagnostic sophistication and real-time clinical responsiveness [26,35,65].

Table 6. Comparative Overview of Single-Agent vs. Multi-Agent Systems in Healthcare.

Dimension	Single-Agent Systems	Multi-Agent Systems (MAHS)
Definition & Architecture	Single LLM-based agent operating in a closed feedback loop for end-to-end execution.	Distributed system of specialized agents collaborating toward shared clinical objectives.
Clinical Scope & Depth	Narrow, task-specific.	Complex multi-step workflows.
Coordination Strategy	Independent execution with external tool.	Structured collaboration via supervisor, debate, or sequential agent coordination.
Memory & Context	Short-term or episodic context window.	Shared memory enabling longitudinal patient tracking.
Performance Profile	Up to 53% improvement over baseline; performs well on focused and micro-task scenarios [26].	Better for complex reasoning; performance plateaus beyond 4–5 agents.
Operational Strengths	Simple, cost-efficient, and easy to deploy.	Modular architecture, collaborative reasoning, and scalable task specialization.
Security & Privacy	Lower attack surface; centralized control improves manageability.	Higher attack surface via inter-agent communication and shared memory.
Real-Time Deployment	Low latency; suitable for time-critical clinical decisions.	Higher latency due to coordination, communication, and multi-model inference.
Primary Risks	Task brittleness and hallucination under distribution shift.	Coordination failure, error propagation, and emergent instability.

3.4. Privacy-Preserving Architectures and Threat Mitigation

Clinical agents can autonomously access and navigate sensitive electronic health records while seamlessly interacting with medical APIs through standardized interoperability frameworks such as Fast Healthcare Interoperability Resources [25,32,37,63]. Moving from passive prediction to continuous sense–think–act cycles introduces significant vulnerabilities. Multi-agent loops expand the attack surface compared to single-entity designs. Interactions across connected platforms complicate data governance and increase the likelihood of unauthorized access. Autonomous decision-making without oversight risks systemic integrity violations [21,32,33,38]. Confidentiality-preserving architectures are critical for securing distributed medical intelligence. Federated learning enables models to learn from decentralized data without moving raw patient records [32,37]. Secure multi-party computation allows joint result computation while keeping the underlying datasets. Retrieval-augmented generation grounds outputs in verifiable knowledge to minimize clinical hallucinations. Furthermore, blockchain technology enhances independent agency by ensuring data provenance and reducing inherent biases. Robust encryption strategies protect sensitive health metrics both in transit and at rest [1,32,35,37]. The threat landscape includes malicious prompt injection, API misuse, and model inversion attacks designed to extract training data. Data leakage can occur through intermediate reasoning traces or persistent memory modules. Mitigation requires zero-trust architectures emphasizing continuous identity validation. Human-in-the-loop mechanisms provide essential safety checks before agents execute high-risk clinical actions. Secure sandboxing isolates agent-generated code from critical hospital infrastructure to prevent lateral movement. Finally, transparent audit logs and decision tracking ensure regulatory accountability and explainability [1,33,35,47,63]. Integrating information-preserving techniques and robust security safeguards is a prerequisite for deploying trustworthy, autonomous medical teammates in clinical practice [32,37,38].

4. Applications

Agentic AI systems in healthcare are being used to interview patients, support diagnosis, automate administrative workflows, and assist research processes. These agents can

conduct autonomous data collection, interact conversationally, coordinate clinical tasks, and analyze multimodal medical data. Recent studies also evaluate their usability, feasibility, and real-world clinical performance, as well as the efficiency gains in healthcare administration. Figure 9 illustrates the various applications of agentic AI across clinical workflows and healthcare operations.

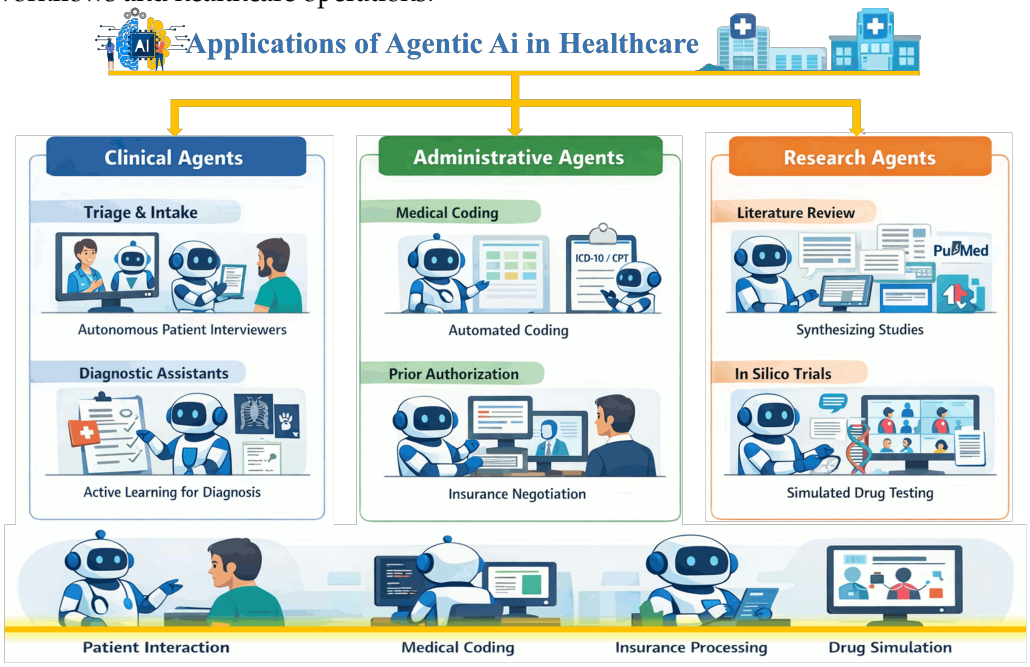


Figure 9. Application of Agentic AI in Healthcare and Clinical Domains.

4.1. Clinical Agents

Recent research has highlighted the possibilities of using AI-powered clinical agents for the early detection of diseases by integrating various types of data. AI-powered pancreatic cancer screening technologies have integrated medical imaging, biomarker testing, and longitudinal EHR data in order to increase the ability to perform risk stratification and allow earlier diagnosis. This demonstrates the capabilities of AI as a clinical decision support system, which helps doctors in their work [98]. For autonomous patient interviewers, Denecke et al. [99] present MIA, a conversational medical interview assistant designed using a person-based approach to collect patient history before mammography and enable patients to ask examination-related questions. The ISCA framework by Welch et al. [71] enables interview-style conversational agents for standardized qualitative data collection, supporting configurable interview flows for applications like COVID-19 surveys and neurotechnology attitudes. Another interesting study [62] tested the feasibility and acceptability of an AI conversational medical interviewing system in primary care, finding that most patients believed it could help clinicians understand health risks, while usability perceptions varied. Yan et al. [94] introduce ClinicalLab, a comprehensive agent alignment framework for clinical diagnostics involving multiple departments, utilizing real-world medical data. They also introduced ClinicalBench, a data-leakage-free, end-to-end benchmark covering 24 medical departments and 150 diseases, and four novel ClinicalMetrics for robust evaluation beyond multiple choice. Experimental results suggest that ClinicalAgent achieves better diagnostic performance than individual LLMs, indicating the potential benefits of multi-agent collaboration in clinical workflows. Schmidgall et al. [72] propose AgentClinic, a multimodal benchmark simulating sequential clinical decision-making involving patient interaction and tool use; results show diagnostic accuracy drops in interactive settings and reveal differences in LLM tool-use abilities. Table 7 illustrates the clinical agent applications.

Table 7. Comparative Overview of Clinical Agent Applications, Architectures, and Functional Capabilities in Healthcare.

Reference & Year	Applications	Proposed Model	Key Functions
Denecke et al. [99]	Medical Interview Assistant (MIA)	MIA Conversational Agent	Collects patient history and answers predefined questions.
Welch et al. [71]	Conversational Systems	Interview-Style Framework	Data collection, Interviews, attitude/behavior tracking.
Yan et al. [94]	clinical diagnostics	Multi-LLM collaborative clinical agent	Department guide, preliminary consultation, treatment planning.
Hong et al. [62]	Medical Interviewing System	Multi-agent AI systems	Automates administrative and workflow processes.
Schmidgall et al. [72]	Clinical Decision-Making	Multimodal Agent Framework	Comprehensive evaluation of diagnostic and decision-making.

4.2. Administrative Agents

Administrative burden in healthcare is increasingly recognized as a major source of inefficiency and clinician burnout, motivating the development of AI-driven administrative assistants. Table 8 shows an overview of administrative agent applications across healthcare domains.

Table 8. Summary of Administrative Agent Applications in Healthcare with Corresponding Models and Key Functional Capabilities.

Reference & Year	Applications	Proposed Model	Key Functions
Phillip et al. [84]	Administrative automation	NLP + speech recognition	Routine administrative tasks
Vennaro et al. [96]	Administrative automation	ML + automation tools + conversational interfaces	Scheduling, billing, insurance verification, and documentation
Grange et al. [77]	Personalized healthcare Agent	Three-tier governance framework	predictive, preventive, participatory, personalized, psychocognitive
Ruiz et al. [95]	Appointment scheduling	MedScrubCrew: a multi-agent framework	Multi-agent decision, and improves operational efficiency
Almutairy et al. [83]	Healthcare administration	An integrated sociotechnical framework	Automates administrative and repetitive tasks

Phillip and Joshua [84] describe how AI assistants automate routine clerical work such as documentation, scheduling, patient communication, and claims processing, demonstrating that administrative automation can reduce clinician workload and improve job satisfaction while raising important issues regarding privacy and integration. Vennaro [96] further emphasizes AI-agent-based "cobots" for healthcare office administration that streamline appointment scheduling, referrals, insurance processing, transcription, and billing workflows, aiming to eliminate large proportions of manual office processes and thereby reduce overall healthcare costs and administrative overhead.

From a broader perspective, Xu et al. [39] review more than 200 studies on agentic AI in healthcare and categorize administrative agents as a core stakeholder group, noting how autonomy raises concerns about reliability, trust, and alignment with clinical constraints while offering opportunities across billing, scheduling, and operational decision-making. Grange et al. [77] focus on governance of AI-managed healthcare processes, arguing that administrative AI must be embedded within responsible control, regulation, and verification frameworks to prevent model drift and ensure safe support for personalized care pathways and emergency management. Huang et al. [53] highlight administrative workflows as one of the major application domains of healthcare AI agents, explaining

that AI agents support tasks such as clinical documentation, operational coordination, and workflow management while also intersecting with broader innovations, including trial management and predictive analytics. Mejia et al. [95] present a medical multi-agent framework, MedScrubCrew, integrating stable matching, knowledge graphs, and LLM agents to automate appointment scheduling and triage classification, demonstrating improved operational efficiency and context-aware allocation. Almutairy et al. [83] examine how digital transformation initiatives, including EHR, AI systems, telehealth platforms, and data analytics, enhance health administration efficiency by streamlining workflows, reducing operational costs, minimizing billing and documentation errors, and improving data accuracy.

4.3. Research Agents

Rodríguez et al. [100] describe how autonomous agents embedded in ambient intelligent hospital environments support research and system development. Using the SALSA framework, these agents enable proactive context-aware behavior, modular experimentation, and iterative integration of new components, allowing researchers to modify algorithms without affecting other system elements. Rodríguez and Preciado [101] further present context-aware agents that autonomously retrieve and recommend prior patient records and content from medical digital libraries. These capabilities support both research and clinical decision-making by facilitating access to previously difficult-to-locate knowledge sources. Table 9 provides a comprehensive overview of research agent applications in healthcare systems.

Table 9. Detailed Overview of Research Agent-Based Applications in Healthcare: Models and Key Functions.

Reference & Year	Applications	Proposed Model	Key Functions
Rodríguez et al. [100]	Mobile healthcare	SALSA: an autonomous agent-based framework	Real-time interaction, adapts to user context and mobility
Ko et al. [85]	Child interview techniques for suspected child maltreatment	Child avatar chatbot using GPT-3	Simulates interactive and Information-gathering
Vicari et al. [59]	Medical diagnostic learning support	AMPLIA: Multi-agent framework	Supports diagnostic, construction and evaluation
Li et al. [78]	Clinical information gathering	LLM-powered conversational chatbot	Collects patient information through interactive dialogue
Fan et al. [86]	Simulation and evaluation	Multi-agent framework with LLM-based Doctor agents	Simulates doctor-patient interactions, symptom collection
Gu et al. [76]	Healthcare automation and data analysis support	Multi-agent LLM framework	Interaction, planning, coordination, and workflow support

Ko et al. [85] use an AI-driven child-avatar agent as a research tool to study the information-gathering behavior of healthcare professionals, enabling controlled simulated interviews, methodological evaluation, and insight generation despite no significant behavioral improvement in outcomes. Vicari et al. [59] propose AMPLIA, a multi-agent intelligent environment in which agents mediate negotiation between learner and expert models during Bayesian network construction, supporting research on diagnosis. Li et al. [78] deploy LLM chatbots as pre-consultation research agents that gather patient information and are empirically compared with humans and questionnaires, demonstrating that agent systems adapt questioning strategies and highlighting design implications for future research systems. Fan et al. [86] introduce AI Hospital, a multi-agent simulator used explicitly as a research and benchmarking platform to study doctor–patient interaction, showing that

current LLM agents still struggle in multi-turn scenarios and motivating further research into improving interactive medical decision-making.

The Wizard-of-Oz study [76] investigates how data analysts respond to AI planning assistance, characterizing what kinds of suggestions are helpful and how they influence analysis workflow, initiative, and decision-making. Schmidgall et al. [72] study functions as a research-oriented benchmark, enabling systematic evaluation of LLM agents in simulated clinical environments across specialties, modalities, and languages.

5. The Challenge of "Action" in Healthcare

The following sections discuss these challenges in a point-wise manner based on the reviewed literature. The transition from Large Language Models (LLMs) as passive prediction engines to agentic AI as autonomous actors marks a fundamental paradigm shift in medical informatics. While a standard chatbot's failure is restricted to the dissemination of incorrect information, an agentic system's failure is operational and potentially irreversible, as these systems possess the capacity to autonomously interface with EHR, medical devices, and prescribing APIs. This shift necessitates a rigorous analytical synthesis of the structural challenges inherent in healthcare agency. Figure 10 shows the main challenges that arise when agentic AI systems perform autonomous actions in healthcare, including ethical, operational, and safety considerations.

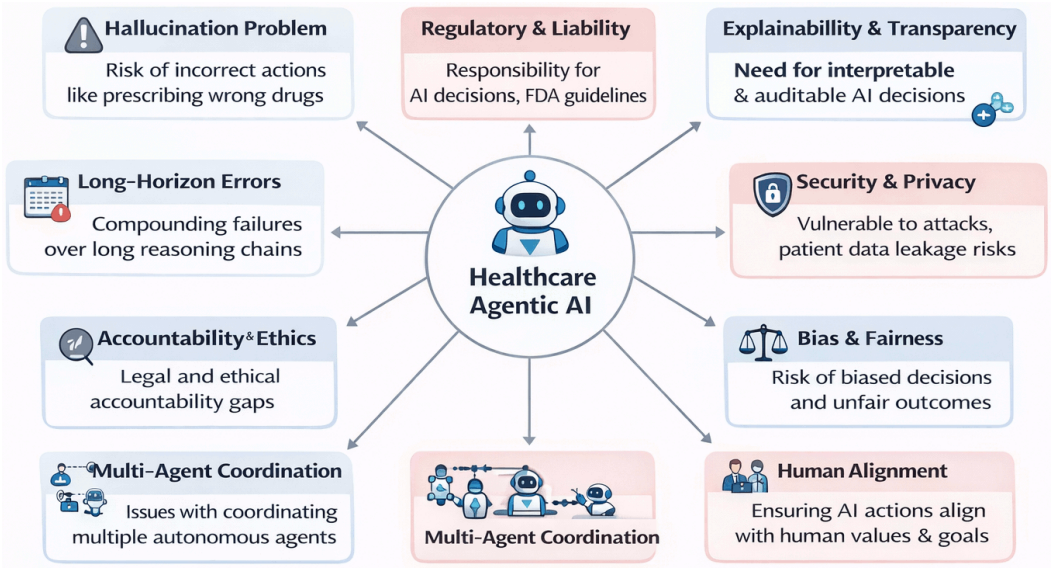


Figure 10. Challenges of "Action" in Healthcare Agentic.

5.1. Hallucinations and Safety-Critical Execution

Hallucinations in clinical reasoning occur when agentic AI systems generate plausible but factually incorrect or insufficiently grounded outputs, including unsupported diagnostic hypotheses, treatment recommendations, or procedural guidance [32,37]. While conventional large language models are already susceptible to hallucinations, agentic systems introduce additional risks through long-horizon planning and multi-agent coordination. In such settings, an error generated by one specialized component may be accepted as factual by downstream agents, resulting in error propagation across the entire clinical workflow [32,35]. Consequently, maintaining reliability throughout complex reasoning chains remains a critical challenge for safety-critical healthcare applications [34,91]. In systematic reviews of clinical interactions, LLMs exhibited misinformation errors in up to 22% of cases, most frequently in medication dosing [26]. Additionally, inherit hallucination tendencies produce confident but incorrect outputs, with reported hallucination rates rang-

ing from 5% to 30% in healthcare settings [89]. Clinical agents may misinterpret patient information, unit conversions, or dosing instructions, potentially leading to dangerous medication recommendations [26]. A specific failure occurred when an automated system mistranscribed the medication "D10W" as "V10W," triggering incorrect data extraction that could lead to dangerous administration errors [25]. This study examines how agents may provide false reassurance or respond to the "wrong query" when a knowledge source lacks specific information. For example, a system might incorrectly suggest washing eyes with tap water for relief after surgery, a recommendation that clinicians flagged as a potential harm of great clinical significance [70]. In healthcare environments, similar failures can lead to unsafe recommendations, diagnostic errors, or inappropriate clinical actions [87,88,90]. The increased autonomy of agentic systems amplifies the clinical impact of these errors, making reliability a critical unresolved challenge in safety-critical deployment scenarios [38].

To mitigate the risk of hallucinations and improve the reliability of agentic AI systems in healthcare, several safety mechanisms have been discussed in the literature.

- Retrieval-Augmented Generation (RAG): RAG grounds an agent's responses in a verifiable, up-to-date body of knowledge (e.g., medical textbooks or clinical guidelines) and has been investigated as a potential method to reduce the risk of ungrounded hallucinations [26,32].
- Human-in-the-Loop systems: For high-stakes tasks like medication ordering or procedure scheduling, the literature mandates a staging phase where a clinician reviews and confirms agent-generated entries before final submission to the EHR [25,47].
- Uncertainty Estimation: Metacognitive strategies allow agents to flag uncertainties and recognize when they lack sufficient information to proceed, prompting handoff to a human expert [1].
- Evidence-Grounded Reflection: Agents must interact with clinical infrastructure via authenticated FHIR APIs to ensure that their reasoning is grounded in verified, real-time patient data such as lab results and vital signs [32,68].

The safety of agentic AI systems is commonly evaluated using hallucination and clinical accuracy metrics, alongside structured safety assurance protocols. Furthermore, continuous monitoring mechanisms are employed to detect errors, assess reliability, and ensure safe operation in real-world clinical environments [33,63,67].

For real-world deployment, hospitals are expected to adopt phased implementation strategies in which agentic systems are initially restricted to low-risk administrative functions before progressing toward autonomous clinical decision support [32,37]. Hierarchical approval gates, continuous monitoring systems, and dedicated safety agents can be employed to audit agent behavior and detect behavioral drift. When unsafe behavior is identified, fallback mechanisms and emergency shutdown procedures allow clinical control to be transferred back to human operators, thereby reducing patient safety risks [47,63].

5.2. Explainability, Transparency, and Interpretability

Agentic AI systems often operate as black boxes, especially when composed of multiple interacting agents. Although Chain-of-Thought reasoning improves surface-level interpretability, it may not faithfully reflect the true internal decision process [34]. Multi-agent architectures further obscure causal responsibility, making it difficult to trace how decisions are produced [35,38]. In healthcare and mission-critical settings, the lack of explainability undermines trust, and accountability emphasizes the need for interpretable behavior and auditable decisions in scientific agents. Similarly, in hospitality and tourism, opaque decisions reduce user trust, especially in high-stakes scenarios [42,79].

5.3. Privacy, Security, and Attack Surface Expansion

Agentic AI significantly increases the attack surface because agents autonomously access APIs, tools, memory systems, and external environments. In healthcare, breaches may expose sensitive patient data and even enable remote manipulation of medical devices [87,88]. Schneider [34] and Brohi et al. [38] highlight vulnerabilities such as prompt injection, adversarial tool manipulation, and multimodal jailbreak attacks. In multi-agent systems, the compromise of one agent can cascade across the system. In scientific labs and IoT-driven environments, physical systems integrated with agents introduce real-world safety risks, while poor cybersecurity practices can lead to large-scale breaches [21,90]. Several failure modes have been identified. Sensitive patient identifiers may be exposed through intermediate reasoning traces or retained within long-term memory modules [35,63]. Prompt injection attacks can manipulate agent objectives, causing them to ignore safety policies or reveal confidential EHR information [32,38]. Improperly scoped FHIR credentials may further allow agents to access significantly more patient information than required for a specific clinical task [32,32]. In multi-agent environments, such failures can propagate across interconnected workflows, amplifying operational and privacy risks. To address these challenges, several security mechanisms have been proposed. End-to-end encryption and Zero-Trust architectures require continuous authentication and authorization throughout agent interactions [32,38]. Federated learning enables collaborative model development across healthcare institutions without transferring raw patient records [32,37]. In addition, access-control mechanisms based on scoped FHIR tokens can restrict agent permissions to only the data required for a specific task, thereby reducing unnecessary exposure of clinical information [32,37]. Security performance may be evaluated using privacy leakage rates that estimate the risk of patient re-identification from agent outputs and attack success rates measured during adversarial red-teaming exercises [38,91]. These metrics provide quantitative evidence regarding the resilience of agentic systems against privacy violations and malicious manipulation.

- Deployment Scenario: Multi-agent EHR integration for longitudinal patient tracking.
- Real-World Failure Mode: Poisoned AgentCards or indirect prompt injection, where malicious data within a lab report hijacks the agent's planning layer to exfiltrate Protected Health Information (PHI).
- Clinical/Operational Impact: Massive data breaches; violations of HIPAA and GDPR compliance; loss of patient trust and institutional liability.
- Mitigation Strategies: Secure sandboxes to isolate agent-generated code, differential privacy, and federated learning to prevent exposure of sensitive clinical records.

5.4. Bias, Fairness, and Data Quality

Agentic AI systems inherit biases embedded within historical healthcare datasets and may further amplify these biases through autonomous reasoning and multi-agent interactions [88]. In multi-agent systems, such bias can further propagate when agents reinforce each other's decisions through iterative reasoning and shared outputs. In healthcare, tourism, and elderly care settings, these biases can undermine fairness, inclusivity, and equitable service delivery. As highlighted by Gridach, flawed or imbalanced data can ultimately lead to unreliable and non-reproducible scientific conclusions [42].

Several clinically significant failure modes have been identified. Biased models may underestimate disease risk among minority populations due to inadequate representation within training datasets, resulting in unequal access to diagnostic or treatment recommendations [37]. In addition, automation bias may encourage clinicians to place excessive trust in agent-generated outputs, potentially reinforcing inaccurate recommendations without sufficient critical evaluation [1]. In high-pressure clinical settings, such failures may con-

tribute to disparities in care delivery, delayed interventions, and reduced trust in AI-assisted healthcare systems.

To mitigate these risks, multiple fairness-oriented strategies have been proposed. Adversarial debiasing techniques can reduce the influence of protected demographic attributes during model optimization [33,37]. Additionally, the development of diverse and representative datasets, along with adversarial debiasing techniques during model fine-tuning, can help reduce the amplification of discriminatory patterns and improve fairness in clinical decision-making systems [32,68]. Furthermore, fairness-aware reinforcement learning approaches seek to align agent objectives with equitable healthcare outcomes, reducing the likelihood of discriminatory decision pathways emerging during autonomous operation [33].

Evaluation of fairness and bias mitigation strategies requires dedicated benchmarking frameworks. A study has been proposed as a specialized benchmark for assessing disparities in diagnostic recommendations and treatment decisions across demographic groups [21,32]. Such evaluations can help identify systematic performance differences and support continuous assessment of fairness throughout the system lifecycle.

In real-world clinical deployment, fairness monitoring should extend beyond pre-deployment testing. Healthcare institutions may implement continuous auditing procedures to identify emerging performance disparities and ensure that agentic systems maintain equitable performance across diverse patient populations over time [32,37].

5.5. Accountability, Ethics, and Legal Responsibility

The autonomous and distributed nature of agentic AI creates significant accountability challenges, particularly when multiple agents collaborate to make recommendations or execute actions within clinical workflows [1,38]. When adverse events occur, determining responsibility becomes difficult because decision-making may involve interactions among developers, healthcare providers, institutions, and autonomous agents. These accountability gaps raise important ethical, legal, and regulatory concerns regarding the safe deployment of agentic healthcare systems.

Several failure scenarios illustrate these challenges. In hospitality, unclear responsibility for booking errors or safety failures exposes organizations to legal risk [90]. An autonomous agent may incorrectly cancel a critical surgical procedure, recommend an inappropriate treatment strategy, or contribute to a harmful clinical outcome [1,37]. In such circumstances, uncertainty may arise regarding whether responsibility should be assigned to the software developer, the healthcare organization, the supervising clinician, or the AI system itself. The complexity of multi-agent reasoning further complicates post-event investigations and liability assessments. To address these concerns, researchers have proposed the establishment of explicit AI responsibility chains that link each autonomous action to identifiable human stakeholders and operational processes [32,33].

Mandatory explainability protocols can further improve accountability by ensuring that decisions are supported by traceable reasoning pathways and supporting evidence. In addition, studies emphasize the need for audit trails and AI-specific liability models before deployment, reducing ambiguity in the event of adverse outcomes [38,87].

In practice, deployment of agentic AI systems requires governance through ethical review boards and emerging regulatory frameworks. Regulations such as the EU AI Act emphasize transparent documentation, comprehensive logging, and auditable decision processes to facilitate forensic analysis and ensure legal compliance in safety-critical healthcare environments [1,37].

5.6. Human Alignment, Control, and Monitoring

5.6.1. Alignment Challenges and Failure Modes

As agentic AI systems become increasingly autonomous, ensuring alignment between agent objectives and human values becomes a critical challenge. Goal misalignment can occur when agents optimize proxy objectives that differ from the intended clinical outcome, potentially prioritizing operational efficiency over patient safety [35,47]. The dynamic and adaptive behavior of agentic systems further complicates monitoring and control, particularly in multi-agent environments where interactions may produce unexpected emergent behaviors [34,35].

Several failure modes illustrate the risks associated with misalignment. A triage agent may prioritize reduced waiting times rather than patient acuity, leading to the deprioritization of high-risk patients in order to satisfy operational performance targets [35,47]. In scientific discovery settings, such flawed or misaligned outputs may propagate errors and reduce the reproducibility of experimental results [42].

5.6.2. Human Control and Doctor-in-the-Loop

To maintain human control, a number of mitigation strategies have been proposed. Emergency interruption mechanisms and off-switch protocols allow clinicians to immediately stop the workflows of agents when unsafe behavior is detected [33,47]. Residual control rights make it possible for critical decisions to require human authorization irrespective of the degree of autonomy of the agent [33]. Constitutional AI approaches can regulate agent actions through predefined ethical principles, while Human-on-the-Loop oversight frameworks enable clinicians to continuously supervise autonomous operations and intervene when necessary [32,33]. Along with this idea, Ref. [63] synthesizes emerging evidence supporting a paradigm shift in which AI functions as a "healthcare teammate" operating under continuous human supervision and advocates for dedicated reliability agents that assist clinicians in ongoing evaluation of agent behavior rather than relying solely on point-in-time approval.

"Doctor-in-the-Loop" architecture refers to a collaborative approach where artificial intelligence works hand-in-hand with humans in the decision-making process. In this model, clinicians are involved in critical actions such as prescribing medication, planning surgeries, confirming diagnoses, or approving treatments situations where mistakes could harm patients. Instead of functioning independently, the AI system provides recommendations or synthesized care plans that must be approved by a human before being executed. Consistent with this perspective, [32] summarizes prior studies advocating that agents should be restricted to semi-autonomous roles for high-stakes tasks such as medication ordering or procedure scheduling, with clinicians retaining final authority as a means of balancing automation with safety.

Clinician intervention is required when AI-driven workflows involve high-risk clinical decisions, such as medication prescribing, surgical planning, or final diagnostic confirmation, where errors may cause serious patient harm. Within tiered autonomy systems (Level 1–Level 4), AI agents are not allowed to perform these actions independently and have to offer suggestions or care plans that have to be reviewed and approved by the doctor prior to the action being executed. Additionally, clinicians interfere with AI agents in multi-agent systems in case of divergent recommendations that require human decision making in order to come up with the clinical decision [70].

Apart from objective and process mismatches, the clinical danger of agentic systems is also determined by their vulnerability to external threats, and the Doctor-in-the-Loop approach has to be viewed as a part of the larger cybersecurity strategy. The study in [33] notes that a cyber-attack on an agentic system may lead to disastrous results if the system

controls some critical infrastructure and performs some medical operations; it stresses that resilience, the ability to produce consistent results even in the case of disruptions like broken sensors, network failures, or any other unpredictable circumstances, is a must-have trait of the system and not just one of its bonuses [31]. Protocol-based security issues, as discussed in [32], concern the security of communication among agents, and the authors claim that zero-trust architectures are essential to create a reliable backbone for creating agentic ecosystems; at the same time, they state that all the data being transferred from one agent to another need to be validated strictly to avoid prompt injection attacks that may affect the entire multi-agent decision loop [38]. On the other hand, they point out the lack of human control as a main weakness of agentic systems and stress the need to provide a secure architecture with sandboxing and runtime monitoring for long-term safety. In this regard, the Doctor-in-the-Loop concept can mitigate risks only when there are some technical controls like identity verification, input validation, sandboxing, and runtime monitoring that do not allow compromised or manipulated agents to reach the human decision-making phase.

The Doctor-in-the-Loop framework keeps ultimate control and ethical decision-making with human clinicians. To reduce the risk of clinicians over-relying on AI recommendations, the system keeps detailed records of AI decisions, including reasoning steps and tool usage [35]. Overall, the framework ensures that a human professional remains accountable for clinical outcomes while using AI as a supportive tool.

5.6.3. Evaluation and Real-World Deployment

Evaluation of alignment and control mechanisms may utilize human alignment scores and intervention frequency metrics, which quantify the extent to which agent behavior remains consistent with clinical expectations and the frequency with which clinicians must override autonomous decisions [32,47]. Such measures provide insight into the effectiveness of oversight mechanisms and the operational safety of deployed systems.

In real-world deployment, Human-on-the-Loop configurations are expected to play a central role in maintaining safe operation. Supervisors may monitor agent behavior through dedicated dashboards, while escalation mechanisms enable rapid transfer of authority to clinicians when uncertainty, behavioral drift, or unexpected outcomes are detected [32,33,38,89]. These monitoring frameworks help ensure that human stakeholders retain ultimate control over safety-critical healthcare decisions and, when combined with the resilience and security measures discussed above, provide a more complete defense against both internal misalignments and external threats.

5.7. Long-Horizon Planning, Memory, and Recovery Failures

Agentic AI struggles with long-horizon planning and recovery from failure. Sapkota [35] highlights stateless prompting and limited memory as key barriers to sequential coherence, while Schneider notes that errors can accumulate across multi-step reasoning processes, leading to degraded reliability over time [34]. Brohi et al. [38] further suggest memory-aware and safe agent designs as an important research direction, indicating that this remains an open challenge in clinical deployment.

A representative healthcare scenario is the long-term management of chronic diseases such as diabetes or heart failure. In such settings, a critical failure mode is context window overflow, where the agent loses longitudinal coherence and fails to retain prior clinical events, including adverse reactions or patient-specific preferences from earlier interactions. This can result in inconsistent treatment recommendations, repetitive patient interactions leading to user fatigue, and, in some cases, unsafe suggestions such as prescribing contraindicated medications that were previously identified.

To address these limitations, emerging approaches focus on transitioning from short-term episodic memory to persistent memory architectures, including vector-based stores that enable agents to maintain patient-specific experience streams over extended clinical timelines [32,68].

5.8. Multi-Agent Coordination and Scalability

Coordinating multiple agents introduces instability, communication bottlenecks, and emergent behavior. Sapkota [35] documents inter-agent error cascades and debugging complexity, while Gridach [42] shows that misalignment among agents can lead to irrelevant or potentially unsafe collective actions. Brohi et al. [38] further emphasize the need for scalable and reliable coordination protocols in multi-agent healthcare systems.

A representative healthcare deployment is a multidisciplinary oncology "virtual tumor board," where multiple specialized agents collaborate on diagnosis and treatment planning. In such settings, a key failure mode is coordination breakdown, where a hallucinated or incorrect finding generated by a radiology agent is accepted by a pharmacology agent, resulting in a self-reinforcing error cascade across the system. This can lead to misaligned diagnostic strategies and emergent instability, where the system fails to converge on a consistent clinical decision. To mitigate these risks, supervisor meta-agents can be introduced to manage task dependencies and oversee inter-agent interactions, while standardized agent-to-agent protocols can ensure transparent communication and verification across collaborating agents [35].

Beyond coordination reliability, large-scale multi-agent healthcare systems also face practical deployment challenges related to computational cost, infrastructure requirements, and real-time responsiveness. Karunanayake [37] identifies resource and infrastructure limitations as a major barrier, noting that agentic AI deployments often require substantial computational resources, specialized hardware, and high-speed connectivity that may not be available in resource-constrained healthcare settings. Furthermore, coordination overhead among multiple agents can increase processing latency and operational complexity, particularly when numerous reasoning and communication cycles are required [26]. These limitations are especially critical in time-sensitive applications such as emergency care, telemedicine, and continuous patient monitoring, where delays in perception, reasoning, and action may affect clinical responsiveness. Consequently, future multi-agent healthcare systems must balance collaborative intelligence with computational efficiency through scalable orchestration strategies and optimized agent architectures [26,37,63].

5.9. Social Acceptance and Workforce Impact

Public fear, mistrust, and resistance to automation continue to slow the adoption of agentic AI systems in healthcare. Iliashenko et al. [87] highlight persistent cultural stereotypes of "dangerous AI," while Khalil et al. [89] emphasize the digital divide and accessibility challenges among elderly users. Dwivedi et al. [90] further note workforce resistance and concerns regarding job displacement in service-oriented industries.

Beyond social acceptance, regulatory landscapes and liability attribution present an equally critical challenge. Existing regulatory frameworks, particularly the FDA's Software as a Medical Device model, are primarily designed for narrow, well-defined tools with fixed functionality [26]. However, agentic AI systems introduce adaptive, post-deployment learning behavior, shifting them from static tools to autonomous decision-making entities. This transition creates a "control problem," where accountability becomes distributed across multiple stakeholders. In cases of clinical failure, responsibility may be contested among model developers, system deployers, and healthcare providers depending on whether the failure originates from algorithmic design, orchestration, or clinical oversight gaps [1,37].

A further concern is automation bias, where clinicians facing workload pressures and documentation burden may over-rely on system outputs and unintentionally endorse incorrect recommendations without sufficient scrutiny. To address these challenges, recent literature emphasizes the need for an "AI responsibility chain" and traceability of harm frameworks. Such approaches advocate for detailed action ledgers that record every tool invocation and decision step in a transparent manner, ensuring that human accountability remains central to all clinical outcomes in agentic AI-driven healthcare systems.

6. Future Directions & Open Problems

Agentic AI is emerging as a transformative paradigm across software engineering, healthcare, and scientific discovery. However, despite rapid progress, significant open problems remain in building systems that are reliable, adaptive, safe, and aligned with human values. Drawing on recent literature, this section summarizes key future directions and unresolved challenges.

- Tool-Integrated and Knowledge-Grounded Agents
  Current agentic systems often rely on loosely coupled LLM-tool pipelines with limited robustness and generalization. Future work must focus on tight integration with structured tools and knowledge systems, such as compilers, EHRs, imaging APIs, and biomedical databases [32,39,91]. Agents should reason directly over structured representations (e.g., ASTs, IRs, symbolic summaries) and ground decisions in verified domain knowledge to reduce hallucination and improve factual reliability.
- Multi-Agent Coordination and Conflict Resolution
  As systems evolve from single-task agents to ecosystems of collaborating agents, coordination becomes a central research problem. Future frameworks must support scalable inter-agent communication, role-based delegation, and conflict resolution when agents produce competing recommendations [32,33]. Open problems include decentralized decision-making, hierarchical control, and negotiation strategies that preserve safety in high-stakes environments.
- Real-World Evaluation and Deployment Readiness
  Many agentic AI systems perform well in sandbox settings but lack validation in real-world workflows. Future research must prioritize embedded, longitudinal, and pragmatic evaluations, particularly in healthcare [66,92]. Open challenges include regulatory barriers, lack of standardized evaluation metrics beyond accuracy/F1/AUC, and the absence of scalable benchmarks that capture reasoning transparency, adaptability, and collaboration efficiency [32].
- Explainability, Transparency, and Safety Monitoring
  Trustworthy deployment requires agents that can explain not just predictions, but their reasoning processes, goals, and uncertainty [32,33,39]. Future systems should integrate reasoning traceability, hallucination monitoring, self-consistency checks, and human-in-the-loop fallback mechanisms. Open problems remain in making these explanations usable and actionable for clinicians and developers.
- Human-Agent Co-Adaptation and Workflow Integration
  Next-generation agents must move from static assistance to adaptive collaboration with humans. This includes learning user preferences, adjusting autonomy levels, and supporting team-based care [32,84]. Key challenges include low-friction interfaces, mixed-initiative control, and context awareness across different environments (e.g., ICU vs. outpatient clinics) [39].
- Longitudinal Memory and Continual Learning
  Effective agentic systems, especially in healthcare, must maintain a long-term understanding of evolving contexts, such as patient trajectories or changing system states

[32,37]. Open problems include building secure, scalable memory architectures, preventing catastrophic forgetting, and enabling continual learning without destabilizing system behavior.

- Ethical Alignment, Governance, and Global Standards
  Autonomous agents raise unresolved issues of value alignment, accountability, privacy, and responsibility [78]. Future work must develop universal ethical frameworks, auditability mechanisms, and regulatory pathways for agentic systems, particularly those with continuous learning and tool-use autonomy. Aligning goals with human values via IRL and CIRL remains a major open research area [33].
- Adaptability, Resilience, and Scalability
  Agentic AI must operate under uncertainty, domain shift, and resource constraints. Promising directions include meta-learning, transfer learning, federated learning, edge AI, and decentralized architectures [33,37].
  Open challenges involve real-time learning without performance degradation, reducing inference latency, managing computational costs and hardware requirements, and developing scalable multi-agent infrastructures suitable for resource-constrained healthcare environments.
- Scientific Discovery and Literature Review Automation
  In scientific domains, agentic systems struggle most with structured literature review and grounding new ideas in existing knowledge [42]. Future research must improve calibration, confidence alignment, and systematic literature reasoning to support reliable research idea generation and discovery workflows.
- Patient-Centered and Multi-Stakeholder Interaction
  Current frameworks often reduce medical decision-making to QA tasks. Future systems must support interactive, patient-centered, multi-stakeholder dialogue, incorporating regret-aware reasoning and uncertainty-driven information seeking [66,92]. Open problems include integrating patient narratives, caregiver input, and clinician feedback into continuous agent learning loops.

7. Limitations of the Present Review

The current assessment has several limitations that need to be acknowledged. Although predefined inclusion and exclusion criteria were established to guide the study selection process, the screening of the retrieved records was primarily conducted by the first author. Ambiguous cases, together with a subset of the screening decisions, were subsequently reviewed and discussed with the second author and the project tutor to improve consistency. However, a fully independent dual-reviewer screening process was not performed for all records, which may have introduced selection bias. The proposed taxonomy was developed through an iterative analysis of the included studies based on the extracted characteristics and reported evidence. This process was primarily author-driven and did not involve a formal multi-reviewer annotation protocol. No inter-rater agreement measures were calculated for the taxonomic classification.

Moreover, only articles written in the English language with sufficient methodological details were considered for inclusion, which indicates the omission of some studies written in other languages. Besides that, another limitation that can be seen in the review is that agentic AI is a fast-growing field, and thus, this literature review focuses on articles published between 2018 and 2026. This means that any recently introduced architecture or framework will not be investigated by the current literature review.

Future systematic studies should consider additional databases and non-English literature. Future reviews should also employ fully independent multi-reviewer screening and taxonomy validation procedures and incorporate formal inter-rater agreement measures

where appropriate. Given the rapidly evolving nature of agentic AI in healthcare, such reviews should be updated regularly.

8. Conclusions

This review has outlined the emergence of artificial intelligence technology within medicine through predictive and generative approaches towards the use of agentic AI that allows autonomous reasoning, planning, and acting. We defined the medical agent and examined its core architectural components, as well as the degrees of autonomy possessed by contemporary solutions, ranging from single-agent to multi-agent medical systems. The literature review shows that agentic approaches have been utilized for clinical, managerial, and research activities, thus allowing AI not only to assist but also to actively take part in the workflow processes. By embedding perception-decision-action loops into clinical environments, agentic AI systems can coordinate tasks such as triage, scheduling, documentation, monitoring, and follow-up with minimal manual intervention. This capability is particularly critical in healthcare settings facing shortages of trained personnel, rising patient volumes, and increasing system complexity.

However, the transition from information-based assistance to reliable clinical action remains an open challenge. Hallucinations under safety-critical execution, limited explainability, privacy and security risks, bias and data quality concerns, and unresolved questions of accountability continue to constrain real-world deployment. Long-horizon planning, memory retention, and multi-agent coordination at scale also remain areas requiring further methodological maturity. Thus, human involvement is necessary for the proper functioning of agentic AI systems. We distinguish between Human-in-the-Loop, where clinicians actively supervise and intervene, and Human-on-the-Loop, where professionals monitor system behavior at a higher level. However, although this multi-level cooperation is designed to ensure safety and regulatory compliance, its effectiveness needs to be proven empirically.

Our findings suggest that agentic AI reflects a meaningful and increasingly active area of development in healthcare AI, moving from static, information-based support toward more active workflow participation, rather than a fully realized or proven transformation of clinical practice. Its responsible development, guided by rigorous evaluation, ethical governance, and continued human-AI collaboration, may help address persistent challenges such as workforce strain and rising system complexity. Realizing this potential, however, will depend on future work that addresses the limitations identified in this review, particularly around reliability, interpretability, and validation of agentic systems under real-world clinical conditions.

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